STOCK PORTFOLIO TRACKER

Submitted by:

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Aim:
To develop a simple Stock Portfolio Tracker using Python that calculates the total investment value based on user-input stock quantities and predefined stock prices.

Objective:
The objective of this project is to help users track their stock investments by entering stock names and quantities. The program calculates the total investment amount using a predefined dictionary of stock prices and optionally stores the result in a file.

Requirements:

- Python 3.x
- Basic knowledge of dictionaries and file handling

Key Concepts Used:

1. Dictionary
2. Input/Output Operations
3. Basic Arithmetic Operations
4. File Handling (Optional)

Working:
The program uses a predefined dictionary that stores stock names and their

corresponding prices. The user is prompted to enter the stock name and the quantity of shares purchased. The program retrieves the stock price from the dictionary and calculates the investment value by multiplying the stock price by the quantity entered.

The process can be repeated for multiple stocks. The individual investment values are added together to obtain the total portfolio value. Finally, the total investment amount is displayed on the screen and can also be saved to a text file for future reference.

Algorithm:

1. Create a dictionary containing stock names and prices.
2. Initialize total investment value to zero.
3. Ask the user for the number of stocks to track.
4. For each stock:
   a. Read stock name.
   b. Read quantity.
   c. Check if the stock exists in the dictionary.
   d. Calculate investment value.
   e. Add the value to the total investment.
5. Display the total investment value.
6. Save the result to a text file (optional).
7. End the program.

Program:

```
# Stock Portfolio Tracker


# Predefined stock prices

stock_prices = {

    "AAPL": 180,

    "TSLA": 250,

    "GOOGL": 140,
```

```
    "MSFT": 330
}

total_investment = 0

# Number of stocks
n = int(input("Enter the number of stocks: "))

# File to save results
file = open("portfolio.txt", "w")
file.write("Stock Portfolio Report\n")
file.write("----------------------\n")

for i in range(n):
    stock = input("\nEnter stock name: ").upper()
    quantity = int(input("Enter quantity: "))

    if stock in stock_prices:
        investment = stock_prices[stock] * quantity
        total_investment += investment

        print(f"{stock} Investment = ${investment}")

        file.write(f"{stock} - Quantity: {quantity}, Investment: ${investment}\n")
    else:
        print("Stock not found!")
print("\nTotal Portfolio Value = $", total_investment)
file.write(f"\nTotal Portfolio Value = ${total_investment}")
file.close()
```

print("Portfolio details saved in portfolio.txt")

Sample Input:

Number of Stocks: 2

Stock Name: AAPL
Quantity: 5

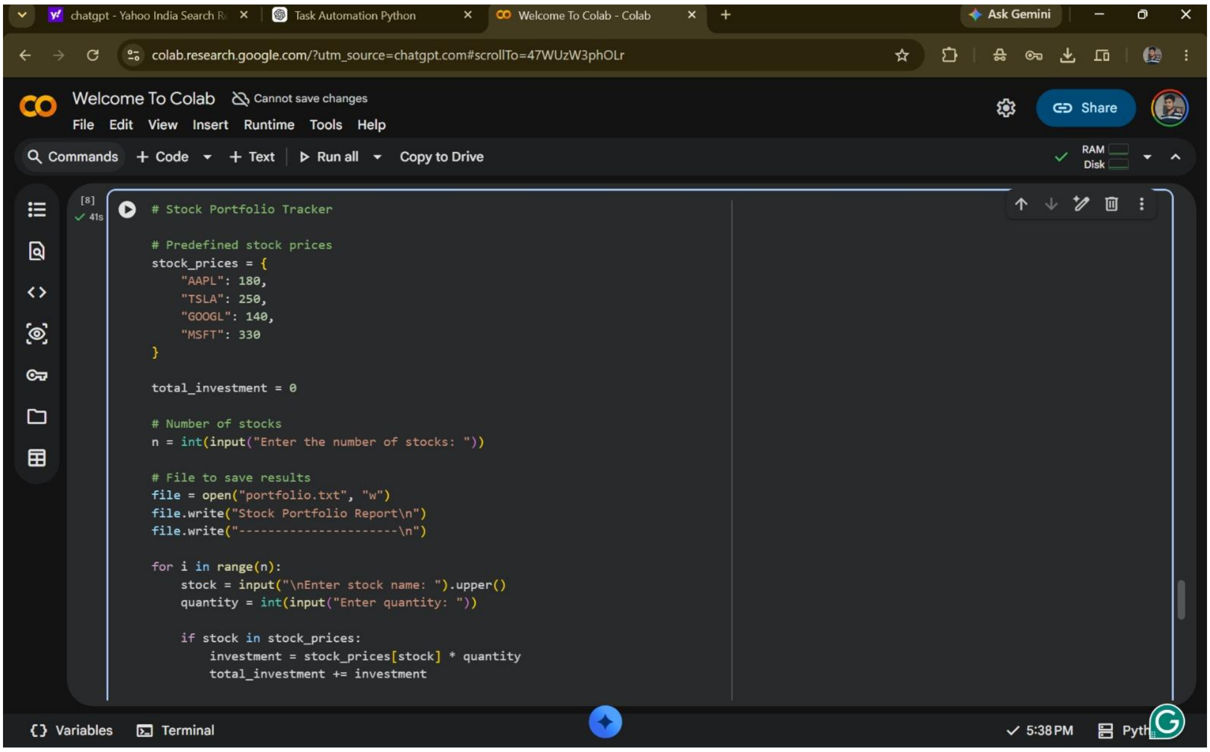
Stock Name: TSLA
Quantity: 3

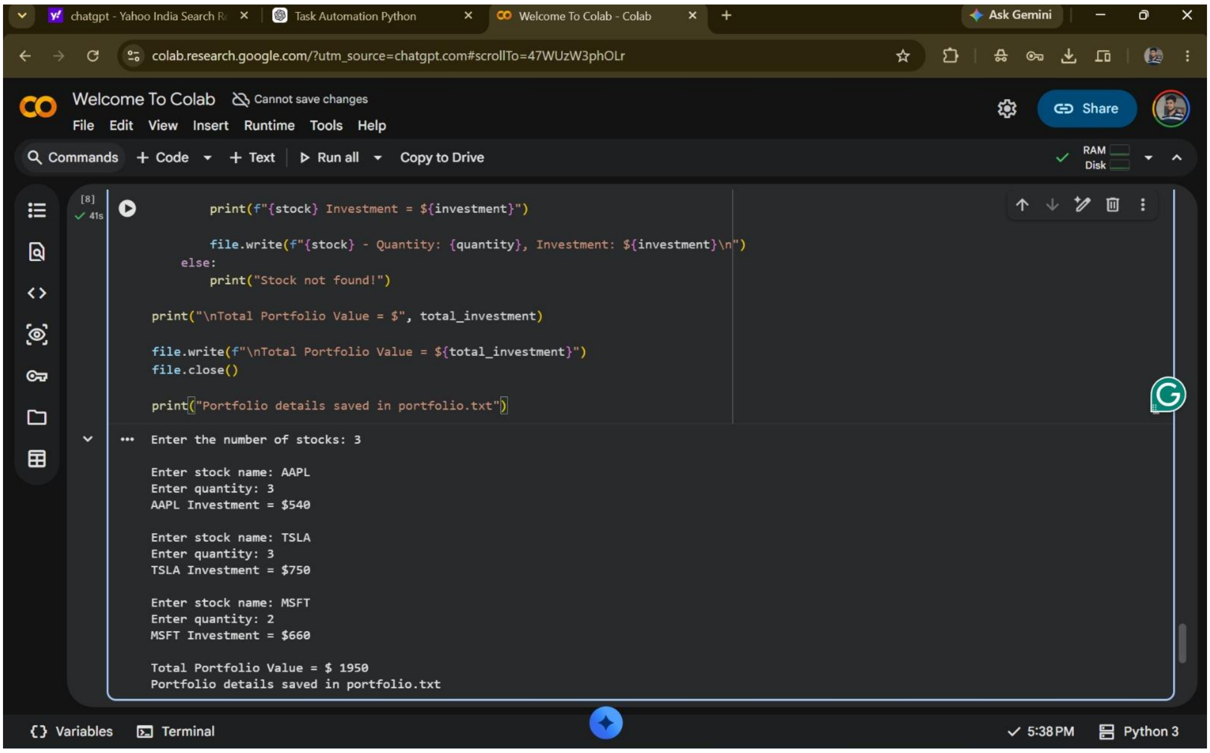
Sample Output:
AAPL Investment = $900

TSLA Investment = $750

Total Portfolio Value = $1650

Output:





Features:

- Accepts stock names and quantities from the user.
- Uses a dictionary to store stock prices.
- Calculates investment value for each stock.
- Computes total portfolio value.
- Displays investment summary.
- Optionally saves results to a text file.
- Simple and user-friendly interface.

Applications:

- Personal investment tracking.
- Financial planning.
- Educational projects for learning Python.
- Portfolio management systems.

Advantages:

- Easy to implement and understand.
- Reduces manual calculation errors.
- Provides quick investment analysis.
- Can be extended with real-time stock prices.

Result:

The Stock Portfolio Tracker was successfully implemented and executed using Python. The program accepted stock names and quantities from the user, calculated the investment value for each stock using predefined stock prices, and computed the total portfolio value. The portfolio details were successfully saved to a text file. Thus, the objective of tracking and calculating stock investments using dictionaries, input/output operations, arithmetic calculations, and file handling was achieved successfully.

Conclusion:

The Stock Portfolio Tracker was successfully developed using Python. The program accepts stock information from the user, calculates the investment value using predefined stock prices, and displays the total portfolio value. The project demonstrates the use of dictionaries, input/output operations, arithmetic calculations, and file handling, thereby achieving the objective of simplifying investment tracking.